

Introduction to Skin Histology

Skin, the largest body organ, comprises three layers—**epidermis** (superficial stratified **squamous epithelium**), dermis (collagen-rich connective tissue), and hypodermis (adipose insulation)—with functions including protection (from desiccation, microbes, UV, mechanical stress), thermoregulation, sensation, metabolism, and sexual attraction. Epidermis regenerates via keratinocyte stem cells, producing keratin barrier; thickness varies (thickest palms/soles). Dermis provides tensile strength via collagen, houses vessels/nerves/receptors; rete ridges (epidermal downgrowths) interlock with dermal papillae (Meissner corpuscles) for shear resistance. Hypodermis offers metabolic energy, insulation, large vessels; eccrine glands border dermis/hypodermis.

Epidermis Layers and Keratinocytes

Epidermis divides into four layers from deep to superficial: stratum basale (cuboidal/columnar stem cells on basement membrane; transit amplifying cells replace migrants), stratum spinosum (polygonal keratinocytes with desmosomes to cytokeratins KRT5/14 basally, KRT1/10 superficially; spiny post-fixation), stratum granulosum (flattened squamous cells with **keratohyaline** granules bundling keratin; lipid vesicles form extracellular hydrophobic barrier), stratum corneum (anucleate apoptotic keratinocytes; keratin sheets + lipids as "brick-and-mortar" barrier against water loss/pathogens). Keratinocytes migrate upward, changing gene expression; desmosomes (desmoglein cadherins) link via intermediate filaments. Mutations in keratins (KRT1/5/10/14) or desmogleins cause blistering diseases.

Layer Cell Features Key Changes/Functions

Stratum Basale Cuboidal/columnar; stem cells Proliferation; melanin uptake

Stratum Spinosum Polygonal; desmosomes Cytokeratin shift; strength

Stratum Granulosum Squamous; granules Keratin bundling; lipid seal

Stratum Corneum Anucleate keratinized Barrier to desiccation/chemicals

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Basement Membrane and Epidermal Attachment

Stratum basale attaches via hemidesmosomes (integrins $\alpha 6 \beta 4$, bullous pemphigoid antigen 2) to basement membrane

epidermis

The word "epidermis" comes from the Ancient Greek words epi- (meaning "outer" or "upon") and derma (meaning "skin"), so its etymological meaning is "outer skin". This term describes the outermost layer of skin in vertebrates and the outer cellular layer of plants.

squamous epithelium

Squamous comes from Latin squama "scale," so it means "scaly" or "scale-like." Epithelium derives from Greek epi "upon" and thēlē "nipple," later generalized to a surface layer. Together it means a surface lining of flat, scale-like cells.

keratohyaline

From Greek keras/kerato- "horn" (used for hard keratin) and hyalos "glass, transparent." Literally "horn-like glassy substance." It refers to the glassy keratin-related granules in certain skin layers.

(**lamina lucida**: integrins; **lamina densa**: type IV collagen, laminins, proteoglycans). Type VII collagen anchors to dermal type I collagen. Electron microscopy shows lamina lucida (light, adhesion molecules) and densa (dark, structural).

Specialized Epidermal Cells

Melanocytes (**stratum basale**; processes to spinosum) produce melanin (eumelanin dark/black, pheomelanin yellow/red from tyrosine via tyrosinase; cysteine influences type) in melanosomes (Golgi-derived; kinesin/myosin transport). Transfer to keratinocytes (phagocytosis) absorbs UV, protects DNA from dimers; dark skin has more eumelanin/retained melanosomes. Langerhans cells (spinosum) are antigen-presenting (capture $\rightarrow$ lymph nodes $\rightarrow$ T-cells).

- Dark skin: abundant basal melanin (brown H&E).
- Light skin: minimal detectable melanin.

Dermis and Hypodermis

Dermis: collagen (tension resistance), vessels, nerves, receptors; **Meissner corpuscles** in papillae (light touch). Hypodermis: adipose (insulation/metabolism), large vessels, Pacinian corpuscles (deep pressure via layered cells/fluid/nerve).

Regional variations: thick epidermis (palms/soles: more layers/prominent rete ridges); thin elsewhere.

Skin Appendages: Hair Follicles

Hair (body-wide; thermoregulation/UV absorption) from epidermal downgrowth: follicle (production site), bulb (stem cells differentiate to concentric keratinized layers), dermal papilla (regulates proliferation). Melanocytes color hair via melanosomes.

Sweat Glands

- Eccrine: Ubiquitous; epidermal downgrowth to coiled secretory tubules (light/dark cells: watery ultrafiltrate + glycoprotein; myoepithelial contraction via nerves), double-layered ducts (NaCl reabsorption $\rightarrow$ hypotonic sweat). Evaporative cooling.

lamina lucida

Latin lamina "thin plate, layer" and lucida from lucidus "bright, clear, shining." Literally "clear thin layer." It describes the transparent zone of the basement membrane.

lamina densa

Latin lamina "thin plate, layer" and densa from densus "thick, dense, crowded." Literally "dense thin layer." It denotes the compact, electron-dense part of the basement membrane.

stratum basale

Latin stratum "spread out layer" and basale from basis "base, foundation." Literally "basal layer" or "layer at the base." It names the deepest germinative layer of the epidermis.

meissner corpuscles

"Corpuscle" is from Latin corpusculum, diminutive of corpus "body," meaning "little body." Meissner is the surname of the German anatomist who described these structures. The term means "Meissner's little bodies," small sensory bodies in the skin.

- Apocrine: Axilla/areola/labia/circumanal; larger, dilated lumen, cuboidal epithelium; protein/lipid/carbohydrate-rich secretion.

Gland Type Location Secretion Structure

Eccrine Throughout skin Hypotonic watery Coiled tubules; myoepithelium

Apocrine Specific regions Thick organic Dilated lumen; cuboidal Histology-of-skin.pdf

Sebaceous Glands

Pear-shaped; attached to hair follicles via **pilosebaceous** canal (near arrector pili). Secrete sebum (oily; moisturizes/waterproofs hair).

Sensory Structures

Epidermis: free nerve endings (temperature/light pressure/pain), Merkel cells (basale; serotonin release to nerves for low-frequency vibration/displacement; clear cytoplasm/granules). Dermis: Meissner corpuscles (papillae tips; light touch: collagen capsule + support cells/nerves). Hypodermis: **Pacinian corpuscles** (deep/coarse pressure: layered cells/collagen/fluid + central nerve).

pilosebaceous

From Latin pilus "hair" and sebum "tallow, grease, fat," plus the adjectival suffix -aceous "pertaining to." Literally "relating to hair and sebum." It refers to the combined unit of hair follicle and sebaceous gland.

pacinian corpuscles

"Corpuscle" again means "little body" from Latin corpusculum < corpus "body." Pacinian comes from the surname of Italian anatomist Filippo Pacini. The term means "Pacini's little bodies," encapsulated pressure receptors in deep tissues.